Ex.No: 1

Study of Microsoft Azure

Introduction

Microsoft Azure is a cloud computing platform launched in 2010 that helps in building, deploying, and managing applications through Microsoft's global data centers. It follows a **Pay-As-You-Go model**, reducing infrastructure cost.

Working of Azure

- Uses **virtualization (hypervisor)** to run multiple virtual machines on a single server
- Supports **Windows & Linux OS**
- Operates through **global data centers** for scalability and performance

Architecture

- Based on **virtualization + software-defined networking (SDN)**
- Provides tools for **development, storage, AI, IoT**
- Uses global infrastructure for high availability

Service Models

1. **IaaS (Infrastructure as a Service)**
   - Virtual Machines, storage, networking
2. **PaaS (Platform as a Service)**
   - App Services, Azure Functions
3. **SaaS (Software as a Service)**
   - Office 365, Azure Active Directory

Key Azure Services

- **Compute** – Virtual Machines, Functions
- **Storage** – Blob, File, Disk
- **Networking** – Virtual Network, Load Balancer
- **Databases** – SQL Database
- **AI & IoT Services**

Use Cases

- Application deployment
- Data storage and databases
- DevOps (CI/CD)
- Identity and access management

Advantages

- Cost-effective (pay only for usage)
- High security and backup support
- Scalable and flexible
- Access from anywhere

Difference between AWS, Google Cloud, and Azure

AWS	Google cloud	Azure
EC2(Elastic Compute Cloud)	Google Compute Engine (GCE)	VHD (Virtual Hard Disk**)**
Fully supports relational and NoSQL databases and Big Data	Fully supports technologies like Big Query, Big Table, Hadoop	upports relational and NoSQL through Windows Azure Table and HDInsight
Per hour — rounded up	Per minute — rounded up	Per minute — rounded up
On-demand, reserved spot	On-demand sustained use	Less "Enterprise-ready"
Simple Storage Service (S3), Elastic Block Storage, Elastic File Storage	Blob Storage, Queue Storage, File Storage, Disk Storage, Data Lake Store	Cloud Storage, Persistent Disk, Transfer Appliance
SageMaker, Lex, Polly, and many more	Cloud Speech AI, Cloud Video Intelligence, Cloud ML Engine	Azure Bot Service, Cognitive Service

Result

The study of Microsoft Azure was successfully completed. Basic concepts such as its architecture, service models (IaaS, PaaS, SaaS), features, and comparison with other cloud platforms like Amazon Web Services and Google Cloud Platform were understood.